

Lakshay Arora

Data Scientist

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PROFESSIONAL EXPERIENCE

Indian Institute of Technology, Delhi

Junior Research Fellow (Computer Vision)

Research Intern

New Delhi, India

May 2025 – Aug 2025

Dec 2024 – Apr 2025

- Built **data science and machine learning workflows** using **Python, OpenCV, MediaPipe, YOLO, Intel RealSense, Scikit-learn, Unity, and MySQL**, converting video/depth streams into structured data for inference and analysis.
- Designed feature extraction pipelines from sequential motion data, enabling pattern analysis, validation, benchmarking, and model-evaluation workflows for evidence-based decision support.
- Tested models and inference outputs on noisy real-world data, translating uncertain signals into documented findings for technical and non-technical stakeholders.
- Mentored **two summer interns** on data collection and project iteration, incorporating feedback from **25 healthy subjects** to improve data quality and analysis.

Maharaja Agrasen Institute of Technology

Research Assistant

New Delhi, India

Sep 2023 – Sep 2024

- Researched **predictive modelling and optimisation-driven ML pipelines** using **Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, and OpenCV** for preprocessing, training, validation, and classification.
- Applied **Particle Swarm Optimisation** and **Genetic Algorithms** for feature selection and model tuning, reducing feature dimensionality by **80%** and improving classification performance by approximately **9%**.
- Built reproducible workflows for model comparison, metric tracking, performance evaluation, technical reporting, and research-driven decision-making.

EDUCATION

King's College London

Master of Science in Data Science

London, United Kingdom

Expected Sep 2026

- **Vice-Chancellor's Award:** £10,000 merit scholarship for top 30 India-domiciled students.
- **Dissertation:** Machine Learning for Cell Behaviour Analysis from 3D Time-Lapse Imaging, modelling movement dynamics and links between visual patterns and gene expression.
- **Relevant Modules:** Neural Networks and Deep Learning; Data Mining; Big Data Technologies; Databases and Information Retrieval; Data Visualisation.

Maharaja Agrasen Institute of Technology

Bachelor of Technology in Computer Science, CGPA: 9.41/10

New Delhi, India

May 2025

TECHNICAL SKILLS

Data Science: Python, SQL, Pandas, NumPy, SciPy, Scikit-learn, statistical modelling, predictive modelling

Machine Learning: linear models, tree-based models, deep learning, TensorFlow/Keras, PyTorch, model evaluation, optimisation

AI Applications: NLP, LLM workflows, prompt engineering concepts, computer vision, OpenCV, Transformers, T5

Communication: business insights, dashboards, technical documentation, stakeholder communication, research communication

PROJECTS

Transformer Text Summarization | *Python, PyTorch, Transformers, T5, FastAPI, Docker, AWS, CI/CD*

- Built an end-to-end **Transformer and T5 NLP workflow** covering data preprocessing, validation, training, inference, and ROUGE/BLEU evaluation.
- Compared model outputs and evaluation metrics to improve summarisation quality, with Dockerised FastAPI inference serving for reproducible experimentation.

Alzheimer's Disease Classification | *Python, TensorFlow, Keras, OpenCV, Machine Learning, Optimisation*

- Developed a **DenseNet-121** predictive classification pipeline covering preprocessing, feature extraction, training, validation, inference, and model comparison.
- Integrated **PSO-GA optimisation** for feature selection and model tuning, reducing feature dimensionality by **80%** and improving classification performance by approximately **9%**.